

CoalGuard

AI-Based Smart Governance, Risk Prediction & Statutory Compliance System for Coal Mines

Problem Statement ID:	SIH26024
Target Ministry / Org:	Ministry of Coal / Coal India Limited (CIL) & DGMS
Category & Theme:	Software Web Development, AI / Machine Learning & Data Analytics
Project Summary:	Offline-first digital safety platform with predictive AI and tamper-evident audit logging for coal mines.

1. Problem Statement

- Indian coal mining operations operate under strict safety laws (Mines Act 1952 and Coal Mines Regulations 2017). However, operations still face critical safety and governance challenges:
- **Paper-Based Registers:** Daily safety logs, Form IV (Accidents), and Form V (Dangerous Occurrences) are manually written, making them prone to physical damage, loss, and unauthorized post-incident alterations.
 - **Underground Communication Void:** Deep seams lack internet connectivity, preventing field sirdars and inspectors from logging hazards and gas readings in real time.
 - **Slow Water Inrush Detection:** Underground aquifer breaches can flood galleries within 15 minutes, but conventional laboratory testing takes 24 to 48 hours to identify the water source.
 - **Opaque AI Decisions:** Traditional machine learning models lack transparency, leading safety officers to mistrust automated alerts.

2. Proposed Solution: CoalGuard

CoalGuard is an industrial-grade, offline-first digital safety governance platform. It replaces manual paper logs with a tamper-evident digital workflow and leverages AI to predict hazards before accidents happen.

Core Capability	How It Works & Practical Impact
Offline-First PWA	Works deep underground without internet using local IndexedDB storage. Automatically synchronizes all logs, photos, and voice notes upon returning to the surface pithead.
Automated PPE Vision	YOLOv8 computer vision analyzes pithead CCTV to detect helmets, reflective vests, and safety boots, auto-generating violation tickets for non-compliance.
Rapid Water Inrush AI	CLSSA-XGBoost classifies breaching water sources across 8 chemical ions in under 1 second with 97.78% precision, replacing 48-hour laboratory delays.
Explainable AI (TreeSHAP)	Provides clear visual factor attributions for every risk prediction, showing exactly which sensor readings drove the risk score.
Multilingual Voice Input	Hands-free voice hazard reporting supporting 10+ Indian languages (Hindi, Bengali, Odia, Marathi, Telugu, English) for grassroots miners.
Tamper-Proof Audit Ledger	SHA-256 cryptographic hash-chaining prevents backdating or retroactive alteration of statutory inspection records.

3. Technical Architecture & Technology Stack

Component	Technologies Used	Function
Frontend (Web & PWA)	React 19, TypeScript, Vite, Tailwind CSS, Workbox, IndexedDB	Responsive, dark/light adaptive dashboard with full offline support and instant sync.
Backend & Database	Supabase, PostgreSQL 15, Row-Level Security, Realtime Channels	Secure relational storage, live alerts via WebSockets, and SHA-256 audit ledger.
AI / ML Engine	FastAPI, PyTorch, YOLOv8, XGBoost, SHAP, Hugging Face	Computer vision for PPE, gradient boosting for risk prediction, and NLP for voice.
Mapping & Reporting	Leaflet GIS, Recharts, jsPDF	Geospatial colliery risk heatmaps and one-click certified DGMS Form V export.

4. Role-Based Portals

- **Colliery Manager Portal:** Monitors active shift hazards, gas sensor feeds, worker strength, and blast zone status.
- **Corporate CIL Dashboard:** Subsidiary-wide risk comparison heatmaps (ECL, BCCL, CCL, SECL, WCL, MCL, NCL) and automated escalation timers.
- **DGMS Regulator Portal:** Statutory compliance inspection view, cryptographic hash verification, and one-click Form V generation.
- **Field Inspectors & Workers:** Offline-capable mobile view for rapid hazard reporting with photos and voice notes.

5. Statutory Compliance Alignment

- **Coal Mines Regulations (CMR 2017) Reg 70:** Daily inspection logs and danger reporting digitized with verification.
- **CMR 2017 Reg 104 & 108:** Noxious gas registers (CO, CH4, CO2) and ventilation velocity monitoring.
- **CMR 2017 Reg 129:** Precaution against water inrush with sub-second hydrochemical classification.
- **Mines Act 1952 Section 23:** Automated digital Form IV reporting for serious incidents.

6. Roadmap & Expected Impact

Phase 1: Pilot (0-3 Months)	Phase 2: Scale (3-6 Months)	Phase 3: Nationwide (6-12 Months)
Deploy in 5 underground mines; calibrate local water ion baselines; pithead PPE camera setup.	Expand to open-cast sites across SECL & MCL; integrate SCADA gas sensors; field training.	Full rollout across all CIL subsidiaries with direct DGMS audit portal federation.

Impact: Eliminates 24-48 hour delays in identifying water inrush sources, prevents register tampering through cryptographic audit logs, and significantly reduces preventable injuries toward a Zero-Harm mining culture.